

Qwen2.5-VL-3B Fine-Tuning Case Study

Fashion-MNIST Baseline and LoRA Comparison

Analysis of repository evaluation artifacts

Chapter 1

Fine-Tuning Results

MNIST is a standard teaching example for neural networks, notably in 3Blue1Brown’s visual introduction to neural networks and throughout much of Michael Nielsen’s open book *Neural Networks and Deep Learning*. The overlap is not coincidental: Nielsen’s book was one of the resources Grant Sanderson used to learn the subject before producing the video. Fashion-MNIST preserves the same compact classification format while replacing digits with clothing images.

1.1 Experimental Question

This case study asks whether parameter-efficient fine-tuning improves Qwen2.5-VL-3B-Instruct on Fashion-MNIST. It compares:

- the unmodified Qwen2.5-VL-3B-Instruct baseline;
- the original Fashion-MNIST LoRA adapter;
- the balanced Fashion-MNIST LoRA adapter.

Each artifact contains the same 50 ordered test examples, enabling paired analysis. Fine-tuning used 100 training and 20 validation examples with seed 42. The adapters use 4-bit QLoRA on the query, key, value, and output projections; the training defaults are two epochs and a learning rate of 10^{-4} .

1.2 Analysis Procedure

The repository comparison schema and telemetry conventions are extended by `comparison/compare_qwen25vl_finetuning.py`. It calculates accuracy, macro and per-class metrics, confusion matrices, paired outcome changes, McNemar’s exact test, a 50,000-resample paired bootstrap, and inference telemetry.

1.3 Overall Classification Performance

Model	Correct	Accuracy	95% CI	Macro recall	Macro F1
Baseline	19/50	0.380	[0.259, 0.518]	0.368	0.342
Original LoRA	19/50	0.380	[0.259, 0.518]	0.349	0.296
Balanced LoRA	24/50	0.480	[0.348, 0.615]	0.455	0.455

Table 1.1: Aggregate Fashion-MNIST performance on the common 50-image test sample. Confidence intervals are Wilson intervals for accuracy.

The original adapter matches the baseline’s 38% accuracy but lowers macro F1 from 0.342 to 0.296. The balanced adapter reaches 48% accuracy and 0.455 macro F1. This is a 10-point accuracy gain and a 16.1% relative reduction in errors.

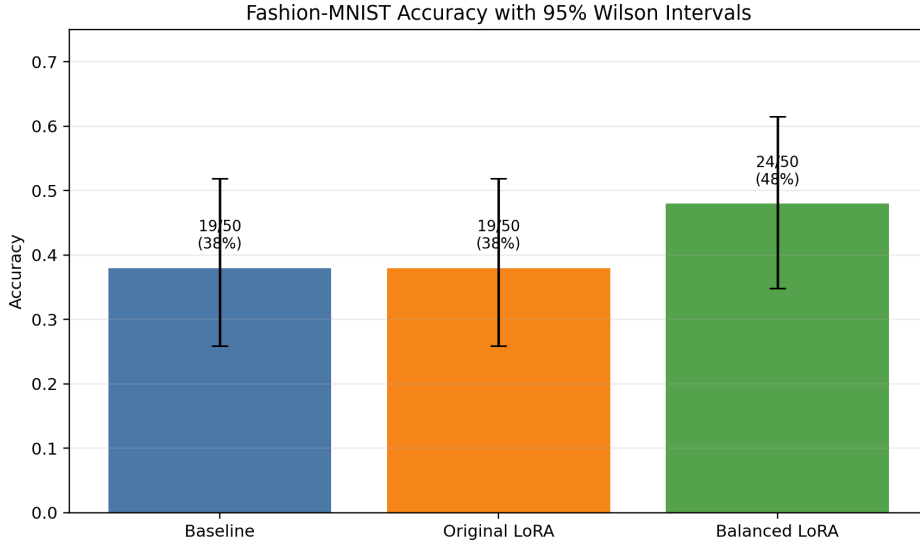


Figure 1.1: Accuracy for the baseline and two LoRA adapters. The wide and overlapping intervals reflect the limited 50-image evaluation sample.

1.4 Paired Statistical Comparison

Candidate	Both right	Fixed	Regressed	Both wrong	McNemar p	Bootstrap 95% CI
Original LoRA	12	7	7	24	1.000	$[-0.14, 0.14]$
Balanced LoRA	14	10	5	21	0.302	$[-0.04, 0.24]$

Table 1.2: Paired comparisons against the baseline. “Fixed” denotes a baseline error corrected by the adapter; “regressed” denotes a baseline success made incorrect by the adapter. Bootstrap intervals describe the change in accuracy.

The original adapter fixes and regresses on seven examples, producing no net gain. The balanced adapter fixes ten and regresses on five. Its effect is promising but inconclusive: McNemar’s $p = 0.302$, and the paired bootstrap 95% interval is $[-0.04, 0.24]$. It beats the baseline in 88.4% of bootstrap resamples, but a larger test is needed for a reliable claim.

1.5 Class-Level Behaviour

Table 1.3: Per-class recall. Class numbers are also used in the confusion-matrix figure.

Index	Class	Baseline	Original LoRA	Balanced LoRA
0	T-shirt/top	0.667	0.000	0.000
1	Trouser	0.000	0.143	0.286
2	Pullover	0.667	0.000	0.167
3	Dress	0.200	0.200	0.200
4	Coat	0.000	0.400	0.400

Index	Class	Baseline	Original LoRA	Balanced LoRA
5	Sandal	0.000	0.000	0.250
6	Shirt	0.400	1.000	1.000
7	Sneaker	1.000	1.000	1.000
8	Bag	0.500	0.750	0.750
9	Ankle boot	0.250	0.000	0.500

The balanced adapter improves six classes but loses ground on T-shirt/top and pullover. Shirt recall rises from 0.40 to 1.00, yet precision is only 0.20 because it predicts *shirt* 25 times for five true shirts. The original adapter is more concentrated still, predicting *shirt* 29 times.



Figure 1.2: Recall by Fashion-MNIST class. The balanced adapter improves several previously weak classes but loses all three T-shirt/top examples and most pullover examples.

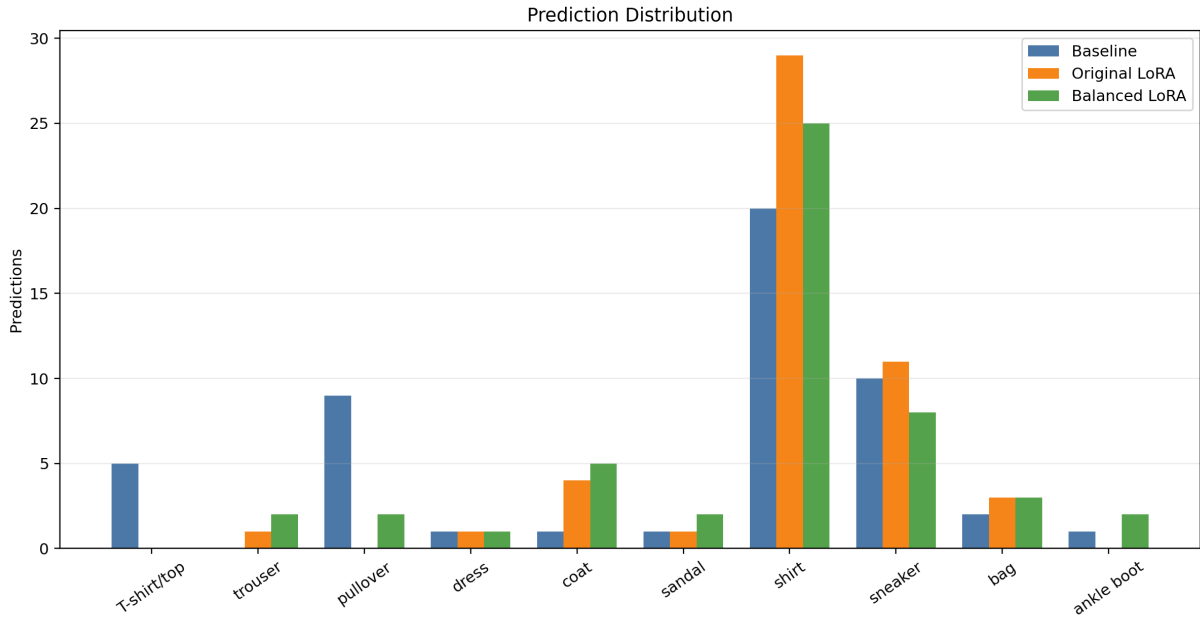


Figure 1.3: Prediction frequencies. Both adapters increase the existing tendency to predict *shirt*; the original adapter exhibits the strongest concentration.

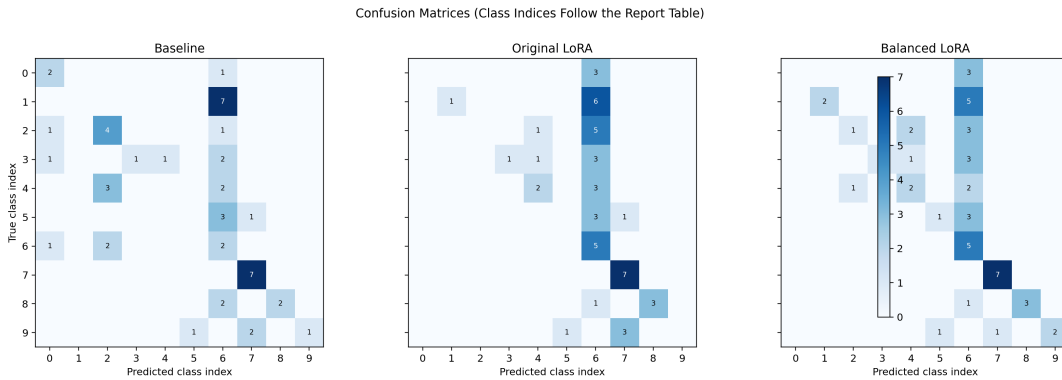


Figure 1.4: Confusion matrices for the common 50-image sample. Rows are reference classes and columns are predicted classes, using the indices in Table ??.

1.6 Inference Efficiency

Model	Total (s)	s/sample	Generation (s)	Samples/s	Peak VRAM (GiB)
Baseline	28.99	0.580	0.547	1.725	7.010
Original LoRA	31.82	0.636	0.611	1.571	7.038
Balanced LoRA	30.39	0.608	0.581	1.645	7.038

Table 1.4: Measured inference telemetry for the 50-image runs. Generation time is the mean per sample.

The balanced adapter adds about 4.8% latency per image and 0.028 GiB peak VRAM while producing five additional correct answers. Timing differences are less controlled than accuracy because the models were run separately.

1.7 Limitations

- Only 50 images were tested, with three to seven examples per class.
- Results come from one training and inference run per adapter.
- The findings are specific to Fashion-MNIST and do not measure general vision-language performance or forgetting.

1.8 Conclusion

The original LoRA provides no improvement. Balanced QLoRA raises observed accuracy from 38% to 48% and macro F1 from 0.342 to 0.455 with little runtime overhead. Because the paired interval includes zero and McNemar’s test is not significant, this is evidence of a promising sample-level gain rather than a conclusive general improvement.